6.3.2 Populations and sustainability

There are many factors that determine the size of a population.

For economic, social and ethical reasons ecosystems may need to be carefully managed.

To support an increasing human population, we need to use biological resources in a sustainable way.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the factors that determine size of a population	To include the significance of limiting factors in determining the carrying capacity of a given environment and the impact of these factors on final population size. *M0.1, M0.2, M0.3, M0.4, M0.5, M1.3, M2.5, M3.1, M3.2* HSW1, HSW2
(b)	interactions between populations	To include predator–prey relationships considering the effects on both predator and prey populations **AND** interspecific and intraspecific competition.
(c)	the reasons for, and differences between, conservation and preservation	To include the economic, social and ethical reasons for conservation of biological resources. HSW7, HSW9, HSW10, HSW12
(d)	how the management of an ecosystem can provide resources in a sustainable way	Examples to include timber production and fishing. HSW12
(e)	the management of environmental resources and the effects of human activities.	To include how ecosystems can be managed to balance the conflict between conservation/preservation and human needs e.g. the Masai Mara region in Kenya and the Terai region of Nepal, peat bogs **AND** the effects of human activities on the animal and plant populations and how these are controlled in environmentally sensitive ecosystems e.g. the Galapagos Islands, Antarctica, Snowdonia National Park, the Lake District. HSW7, HSW12